

VELIKIYE LUKI, Russian Federation

WMO: 264770

Lat: 56.350N

Lon: 30.617E

Elev: 338

StdP: 14.52

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.0	-2.6	-13.8	2.6	-8.0	-7.5	3.7	-1.2	19.6	22.5	17.4	29.0	1.5	100	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	82.8	67.1	79.9	66.0	77.1	64.6	69.9	78.5	68.0	76.7	66.1	74.0	4.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.7	99.6	74.0	64.9	93.2	71.5	63.1	87.6	69.9	34.0	78.2	32.4	76.8	31.0	73.9	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
16.9	14.6	12.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-16.9	87.1	7.8	2.6	-22.5	89.0	-27.0	90.5	-31.4	92.0	-37.0	93.9
			-17.2	72.4	7.6	1.8	-22.7	73.7	-27.2	74.8	-31.5	75.8	-37.0	77.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	21.6	23.8	30.9	44.6	54.1	61.6	64.6	62.0	52.8	42.7	32.7	26.1	
	DBStd	17.45	11.42	11.87	8.38	8.64	8.34	6.83	5.59	5.43	6.92	7.45	9.44	10.79	
	HDD50	4027	880	733	593	211	53	3	0	0	50	245	519	740	
	HDD65	8098	1345	1153	1058	613	348	144	76	126	369	693	968	1205	
	CDD50	1556	0	0	0	48	181	352	453	372	133	17	0	0	
	CDD65	154	0	0	0	0	11	44	64	33	2	0	0	0	
	CDH74	1380	0	0	0	7	124	365	523	328	33	0	0	0	
	CDH80	244	0	0	0	0	18	54	114	57	1	0	0	0	
Wind	WSAvg	5.2	6.0	6.2	6.3	5.4	5.0	4.4	4.1	3.9	4.2	5.3	5.7	6.1	
Precipitation	PrecAvg	23.5	1.3	1.2	1.2	1.3	2.1	2.9	3.2	2.8	2.2	2.0	1.7	1.7	
	PrecMax	32.7	3.3	2.8	2.8	3.8	5.0	7.2	8.4	8.0	6.6	4.3	4.3	2.7	
	PrecMin	17.2	0.2	0.1	0.2	0.1	0.4	1.2	0.7	0.2	0.4	0.2	0.0	0.6	
	PrecStd	3.9	0.6	0.6	0.6	0.9	1.1	1.3	1.5	1.7	1.4	1.0	0.8	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	42.8	56.1	75.3	82.6	84.9	87.4	85.1	78.3	65.8	52.7	47.2	
		MCWB	39.3	38.3	44.3	55.6	65.2	68.0	70.6	66.9	64.2	56.3	50.7	44.7	
	2%	DB	37.8	39.7	49.3	70.2	77.6	81.3	83.3	80.9	72.3	59.4	49.5	42.6	
		MCWB	36.4	36.2	40.8	54.4	62.2	66.1	68.9	66.2	61.9	53.2	47.3	41.2	
	5%	DB	36.1	37.7	44.9	64.7	73.2	78.4	79.6	77.5	67.8	55.8	47.3	40.2	
		MCWB	34.5	35.3	38.6	52.5	59.2	65.0	67.1	64.8	59.2	51.7	45.4	39.2	
	10%	DB	34.6	36.3	41.5	59.7	69.0	75.1	76.3	74.1	63.9	53.4	44.6	37.6	
		MCWB	33.3	34.4	37.4	49.5	57.5	63.7	65.6	62.9	57.0	50.3	43.1	36.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	39.2	46.7	58.5	67.4	70.7	74.9	70.5	65.6	57.9	51.3	45.1
			MCDB	40.5	41.8	53.9	71.0	79.4	80.1	85.1	79.8	74.5	64.0	52.2	46.4
		2%	WB	36.9	36.9	42.3	55.6	64.1	68.7	71.0	67.9	62.8	54.3	47.7	41.3
			MCDB	37.6	38.4	46.2	66.1	73.6	77.8	78.9	77.2	70.8	57.5	49.1	42.4
5%		WB	34.9	35.8	39.9	53.3	61.7	66.8	68.8	66.0	60.5	52.5	45.6	39.2	
		MCDB	35.8	37.3	43.4	62.9	70.2	74.6	76.8	74.6	66.3	55.5	47.2	40.2	
10%		WB	33.5	34.6	37.7	50.8	59.1	64.9	66.6	64.4	58.4	50.3	43.0	36.5	
		MCDB	34.4	36.0	41.1	59.2	66.8	72.7	74.0	71.9	62.7	53.3	44.5	37.4	
Mean Daily Temperature Range	5% DB	MDBR	8.9	11.7	13.5	17.5	19.3	18.0	17.2	19.0	16.3	11.6	7.7	7.1	
		MCDBR	6.5	8.9	17.8	25.9	25.6	22.6	23.3	25.0	23.4	16.2	9.0	6.5	
		MCWBR	6.3	7.1	10.9	12.6	11.6	10.0	10.6	11.5	12.8	10.8	8.0	6.0	
	5% WB	MCDBR	6.3	7.1	13.7	23.9	22.1	19.8	20.6	21.3	21.3	14.8	8.5	6.2	
		MCWBR	6.3	6.3	9.4	12.2	11.4	9.6	10.4	11.1	12.2	10.7	7.9	6.0	
Clear-Sky Solar Irradiance	taub	0.288	0.322	0.347	0.397	0.386	0.376	0.401	0.408	0.369	0.348	0.328	0.288		
	taud	2.146	2.117	2.169	2.228	2.340	2.389	2.337	2.309	2.394	2.383	2.245	2.162		
	Ebn at Noon	186	226	254	259	271	275	264	253	248	220	171	154		
	Edh at Noon	18	28	35	38	36	35	36	35	28	22	17	14		
All-Sky Solar Radiation	RadAvg	158	426	882	1244	1629	1733	1675	1337	855	389	135	93		
	RadStd	36	61	107	148	155	153	161	136	84	81	23	27		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page